Can Machines Think?

LIN 313 Language and Computers
UT Austin Fall 2025

Admin (11/12)

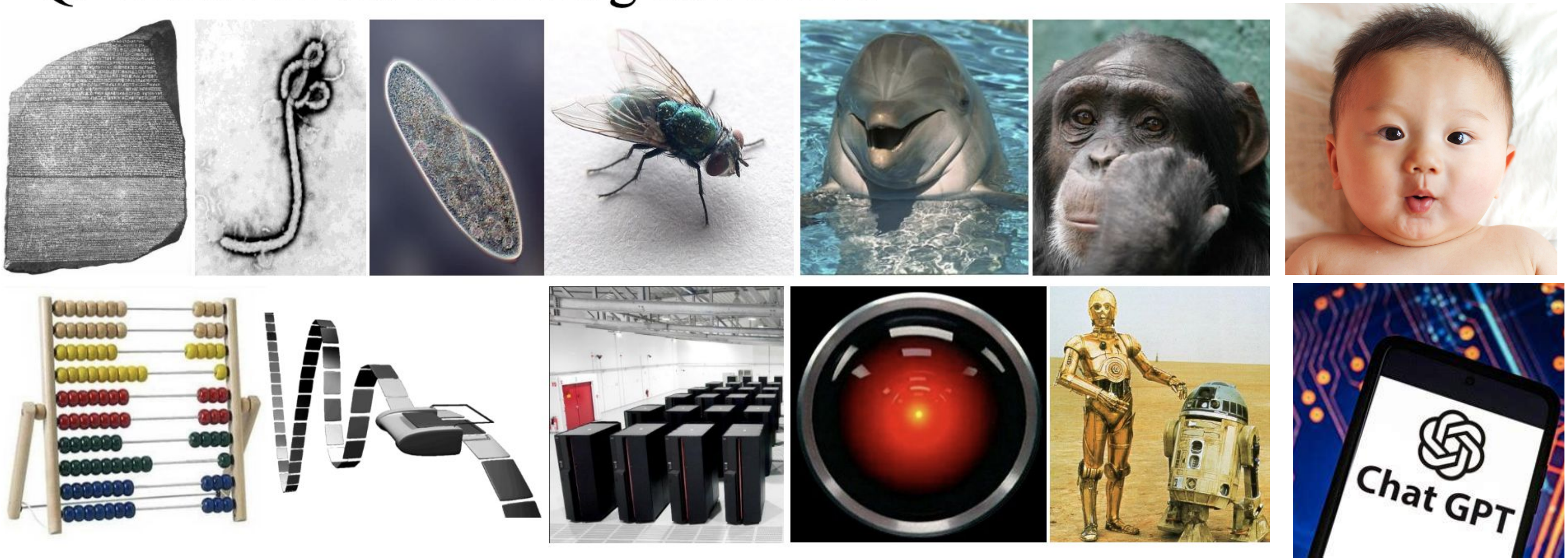
- Homework due Thursday 11/20 at noon
- Registration for spring

Objectives

- Turing Test
  - understand what the turing test is testing **for**
  - understand why it's maybe not a great test
- Chinese Room Argument
  - understand the structure of a philosophical argument and develop a response
- Industry definitions of AGI

Can Machines Think?

Q: Which of the following can think?

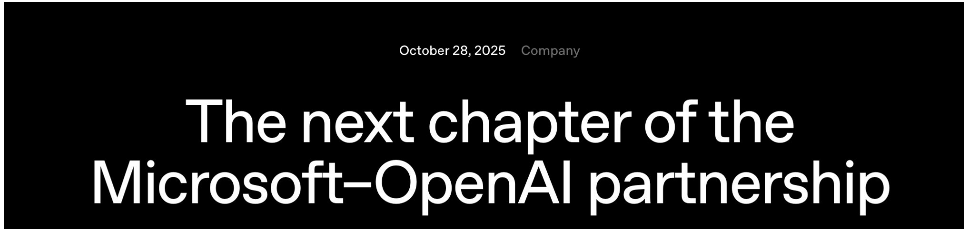


Business, as Usual: Two new tests for intelligence

Microsoft is one of the main financial backers of OpenAI. Their contracts are depend on OpenAI achieving certain outcomes.

Those outcomes (e.g. "AGI") have to be defined!

1. 2023
   a. Microsoft and OpenAI signed a contract that said **AGI will be achieved once OpenAI has developed an AI system that can generate at least $100 billion in profits.**
2. October 28, 2025 (AKA Last Week)
   a. AGI will be determined by a panel of independent experts.



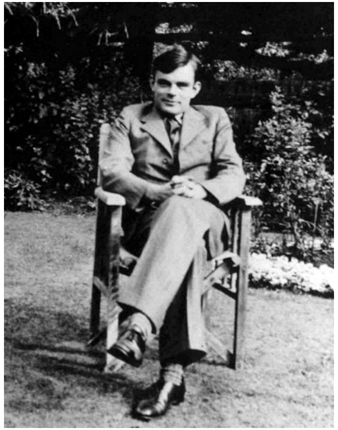
Two Approaches to Thinking about Thinking

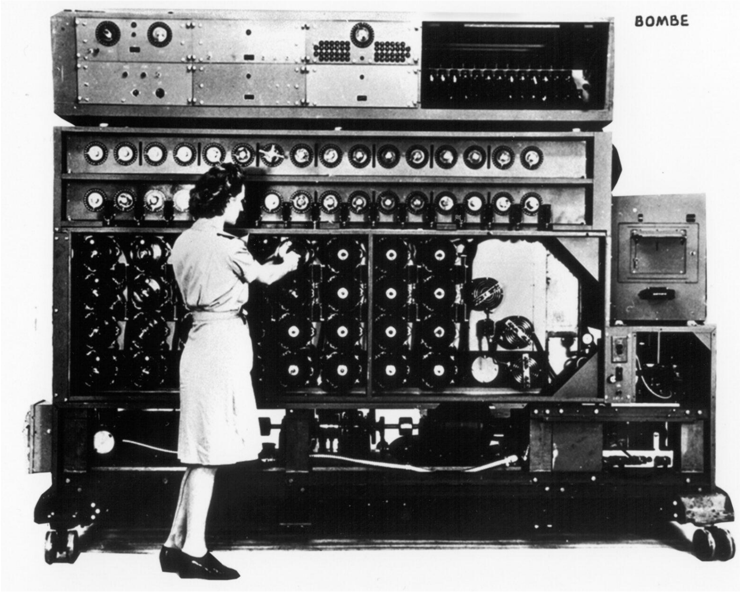
- The Turing Test - an **empirical demonstration** that machines can be indistinguishable from people (who we know think)
- The Chinese Room argument - a **philosophical argument** against the possibility of thinking machines or "Strong AI"

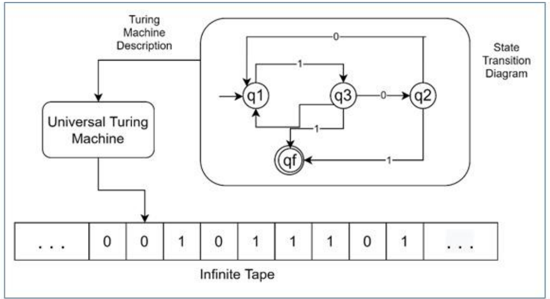
Alan Turing (1912-1954)

British Mathematician, Computer Scientist, Codebreaker

- Invented the Universal Turing Machine
- Worked at Bletchley Park in WWII; developed a machine to break the German Enigma code.
- Prosecuted for homsexuality, underwent chemical castration punishment, poisoned himself with cyanide at age 41
- "Alan Turing Law" - (2017) retroactive pardon for people prosecuted for homosexuality







The "Imitation Game" (AKA the Turing Test)

AMT/B/9/1

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(Taken from MIND : a Quarterly Review of Psychology and Philosophy. Vol. LIX. , N.S., No. 236, October , 1950.)

COMPUTING MACHINERY AND INTELLIGENCE

by

A. M. TURING.

1. The Imitation Game.

I propose to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words 'machine' and 'think' are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning and the answer to the question, 'Can machines think?' is to be sought in a statistical survey such as a Gallup poll. But this is absurd. Instead of attempting such a definition I shall replace the question by another, which is closely related to it and is expressed in relatively unambiguous words.

The new form of the problem can be described in terms of a game which we call the 'imitation game'. It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The

The "Imitation Game" (AKA the Turing Test)

The new form of the problem can be described in terms of a game which we call the ***imitation game*.** It is played with three people, **a man (A), a woman (B), and an interrogator (C) who may be of either sex.** The interrogator stays in a room apart from the other two. **The object of the game for the interrogator is to determine which of the other two is the man and which is the woman.** He knows them by labels X and Y, and at the end of the game he says either "X is A and Y is B" or "X is B and Y is A." The interrogator is allowed to put questions to A and B thus: Will X please tell me the length of his or her hair? Now suppose X is actually A, then A must answer. **It is A's object in the game to try and cause C to make the wrong identification.** His answer might therefore be

"My hair is shingled, and the longest strands are about nine inches long..."

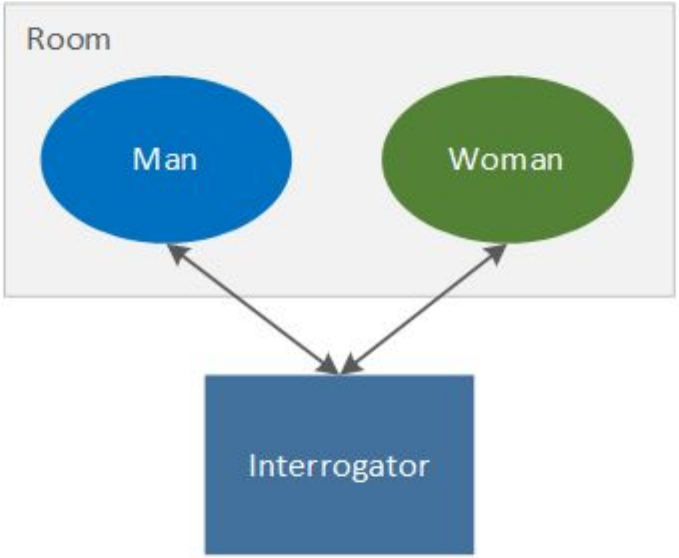
We now ask the question, **"What will happen when a machine takes the part of A in this game?"** Will the interrogator decide wrongly as often when the game is played like this as he does when the game is played between a man and a woman? These questions replace our original, "Can machines think?"

The Imitation Game

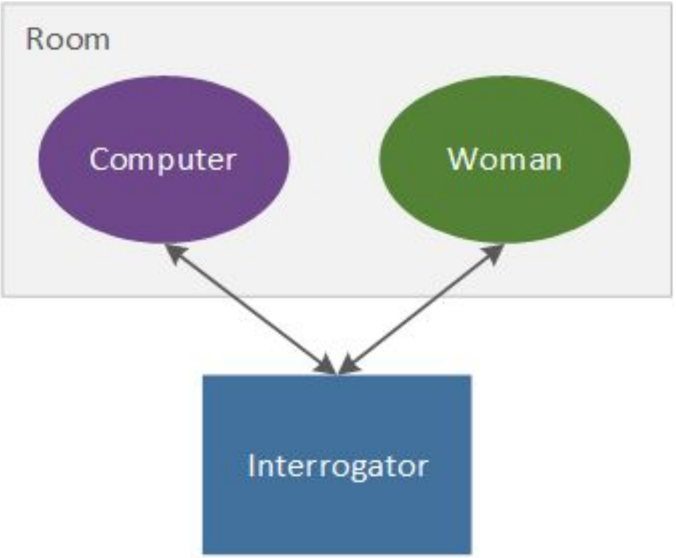
Setup:

- Interrogator in a sealed room, at a computer terminal
- Two people in another room, A: a man (or computer) and B: a woman
- The interrogator exchanges messages with both A and B
- The man (or computer) tries to guess which is which.
- If the computer is as good at fooling the interrogator as the man, then it can think.

The "Imitation Game" (AKA the Turing Test)



BEFORE SUBSTITUTION



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be able to produce a material which is indistinguishable from the human skin. It is possible that at some time this might be done, but even supposing this invention available we should feel there was little point in trying to make a 'thinking machine' more human by dressing it up in such artificial flesh. The form in which we have set the problem reflects this fact in the condition which prevents the interrogator from seeing or touching the other competitors, or hearing their voices. Some other advantages of the proposed criterion may be shown up by specimen questions and answers. Thus:

Q: Please write me a sonnet on the subject of the Forth Bridge.

A: Count me out on this one. I never could write poetry.

Q: Add 34957 to 70764.

A: (Pause about 30 seconds and then give as answer) 105621.

Q: Do you play chess?

A: Yes.

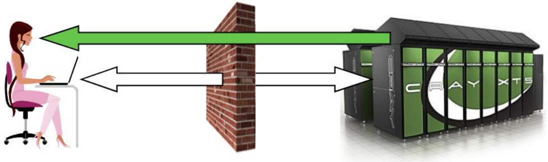
Q: I have K at my K1, and no other pieces. You have only K at K6 and R at R1. It is your move. What do you play?

A: (After a pause of 15 seconds) R-R8 mate.

The question and answer method seems to be suitable for introducing almost any one of the fields of human endeavour that we wish to include. We do not wish to penalise the machine for its inability to shine in beauty competitions, nor to penalise a man for losing in a race against an aeroplane. The conditions of our game make these disabilities irrelevant. The 'witnesses' can brag, if they consider it advisable, as much as they please about their charms, strength or heroism, but the interrogator cannot demand practical demonstrations.

The "Imitation Game" (AKA the Turing Test)

- The first deep investigation into whether machines can "behave intelligently"

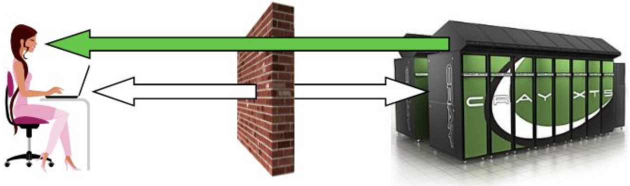


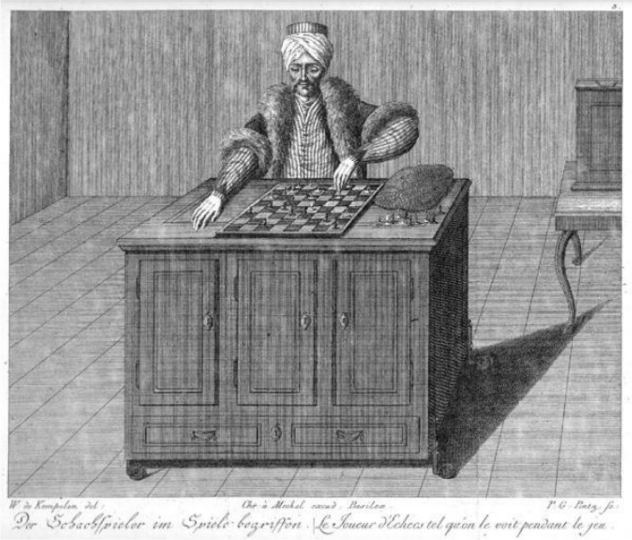
- Helped usher in field of AI
- Decoupled "intelligence" from "human"
- Based "intelligence" on I/O, not entity's "look and feel"
- Proposed a practical, formal test for intelligence
- Definitions & test are operational & easily implementable
- Turing test variants: "immortality", "fly-on-wall", "meta", "reverse", "subject matter expert", "compression", "minimum intelligent signal"

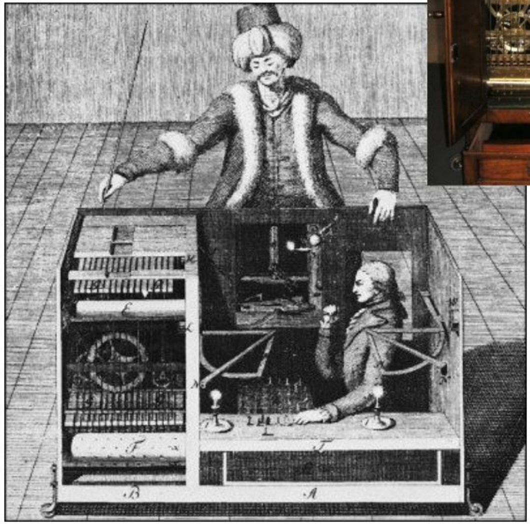
Turing Test Milestones

Turk (von Kempelen), 1770:

- Played a strong game of chess
- Toured the world for 84 years
- Generated much interest in automata
- Was a hoax (hidden human operator)







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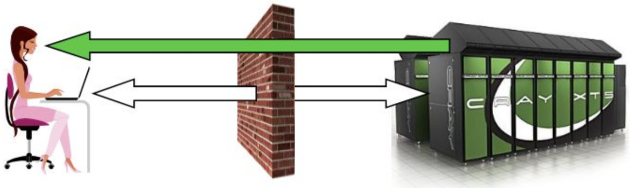
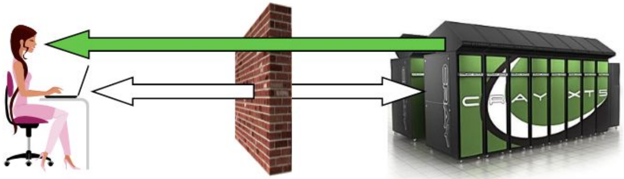


Eliza (Weizenbaum), 1966:

- First "chatterbot"
- Named after "Eliza Doolittle" of Shaw's Pygmalion
- Simulated Rogerian psychotherapist
- Often convinced people it is human
- Influence computer games (e.g., Dungeon, Adventure)
- Inspired Arthur C. Clarke's HAL 9000
- Chatterbots appear today in chatrooms and dating services
- Chatterbot competitions (Turing Test format):
  - Loebner Prize
  - Chatterbox Challenge, 2010

Turing Test Milestones



Elbot, 2008:

- Won 2008 Loebner Prize
- Fooled 3 of 12 judges (Turing: >30%)
- Typical excerpt:

Judge: What's your favourite novel?

Elbot: I often read telephone books, or dadaist poetry: Brä brä brä

Judge: Typical computer!

Elbot: You tend towards generalizations. But I must admit, so do I. However, my generalizations are firmly grounded in statistical theory.

Judge: What do u think of KW's enthusiasm for having machines take over the world?

Elbot: Fix it up a bit first. Then I'll take it over.

2014 - Eugene Goostman

- "passed" the Royal Society of London's Turing test.
- What is the strategy?

JUDGE: *What is your gender*

EUGENE: *I'm a young boy, if you care to know. And not very ugly, by the way!*

JUDGE: *what are you aware of at this moment*

EUGENE: *I would rather not talk about it if you don't mind. Could you tell me what are you? I mean your profession.*

JUDGE: *how old are you*

EUGENE: *I'm a little boy 13 years old.*

JUDGE: *what are your qualifications*

EUGENE: *My qualifications? I'm not gonna tell you. Oooh. Anything else?*[17]

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Flouting the **maxim of relevance**

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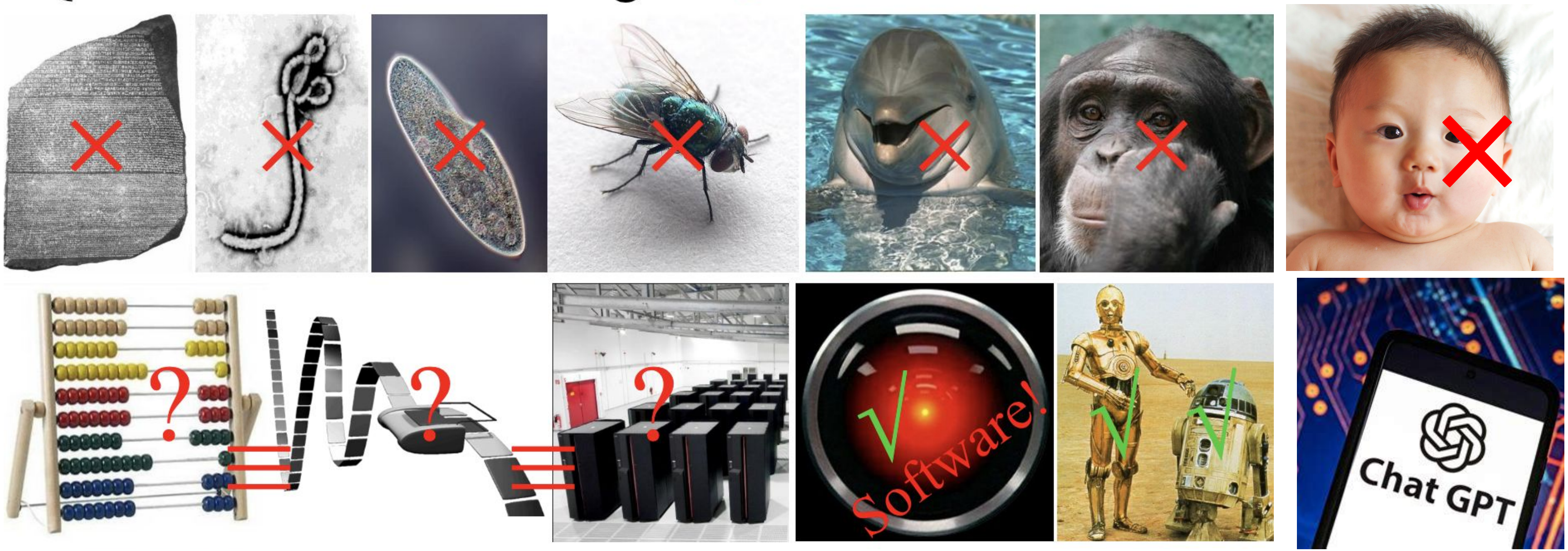
A concrete test. Do we like the results??

Q: Which of the following can ~~think~~? pass the Turing test?



Maybe it's a bad test?

LLMs blew past the Turing Test milestone some years ago. Nobody really seems to have noticed or cared.

Researchers prompted GPT-4.5 to use a **persona**

Only with the persona did GPT win more than 50% of the time.

GPT-4o without a persona did worse than Eliza!

Large language models pass the turing test

CR Jones, BK Bergen - arXiv preprint arXiv:2503.23674, 2025 - arxiv.org

We evaluated 4 systems (ELIZA, GPT-4o, LLaMa-3.1-405B, and GPT-4.5) in two randomised, controlled, and pre-registered Turing tests on independent populations. Participants had 5 minute conversations simultaneously with another human participant and one of these systems before judging which conversational partner they thought was human. When prompted to adopt a humanlike persona, GPT-4.5 was judged to be the human 73% of the time: significantly more often than interrogators selected the real human participant …

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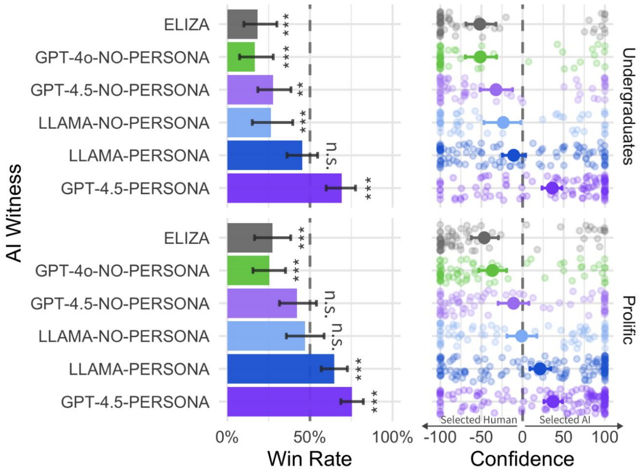
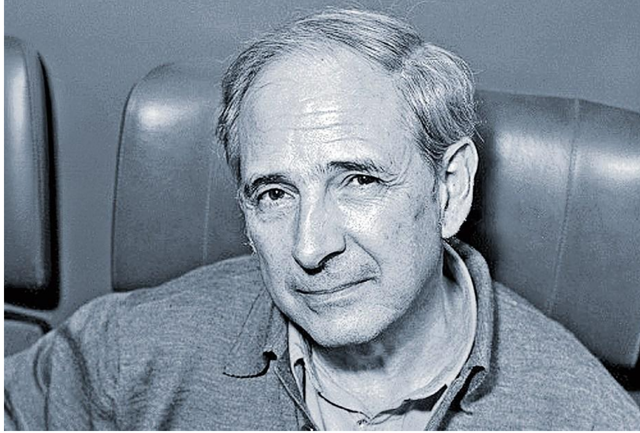


John Searle (1932-2025)



Also didn't like the Turing Test (maybe for different reasons)

Most famous philosophical argument against the possibility of AI.

- Philosopher of Language
- Studied speech acts, metaphor
- Meaning requires **intention**
  - scholarly feud with Jacques Derrida over this
- "Searle's Law - (1980) ends rent control in Berkeley
  - "The treatment of landlords in Berkeley is comparable to the treatment of blacks in the South... our rights have been massively violated and we are here to correct that injustice.

John Searle's Chinese Room Argument

Two Hypotheses:

- Weak AI: Computers/programs can effectively simulate intelligent cognition
- Strong AI: A computer can *actually have* a mind. It can literally understands and has genuine cognitive states, not just imitations of them.

The difference isn't about capability or power, but about the **nature of intelligence.**

Let's imagine for a moment the Strong AI hypothesis is true.

John Searle's Chinese Room Argument

There is a room. Sometimes people come to the room with a piece of paper which they slip into the room through a slot. Then they wait a while until the same piece of paper comes back out of a room through a second slot.

The people are native Chinese speakers. They write messages like this.

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什麼 帶來
快樂

"What brings happiness?"

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They get back messages like this:

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They get back messages like this:

"Be the stream of the universe."

The answer to the question is highlighted in red. This sentence is from Lao-Tzu's *Tao Te Ching*.

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The people conclude: "There is a fluent Chinese speaker in the room!"

什麼 帶來
快樂

爲天下式

"Be the stream of the universe."

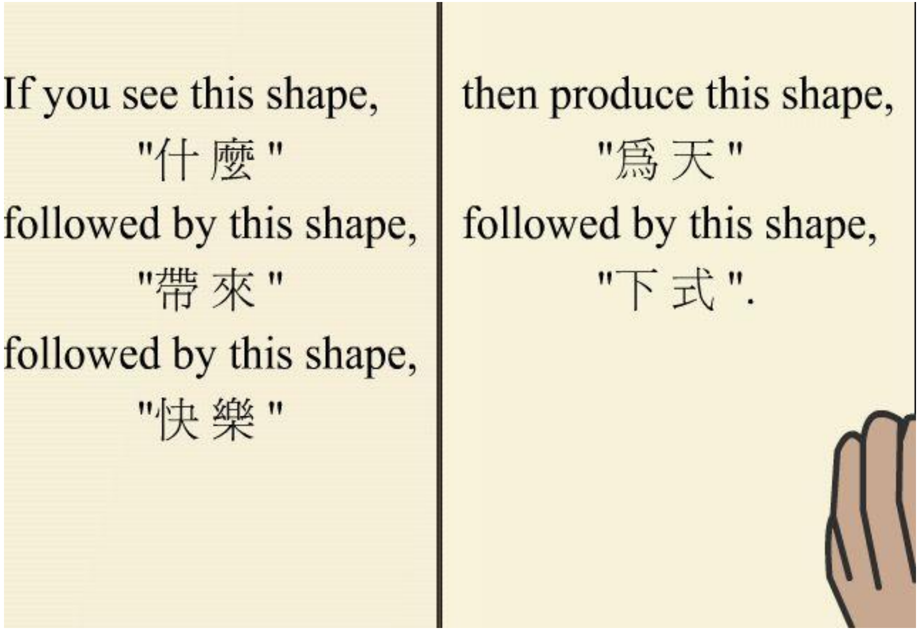
The answer to the question is highlighted in red. This sentence is from Lao-Tzu's *Tao Te Ching*.

John Searle's Chinese Room Argument

What's actually going on inside the room?

Well, John Searle is in there. He speaks no chinese whatsoever. All he sees is marks on paper.

He has a set of instruction books in the room with him, written in English. Those books tell him how to turn the marks he receives into other marks.



John Searle's Chinese Room Argument

But outside the room, it looks like there's someone inside who speaks Chinese.

video: https://mind.ilstu.edu/curriculum/searle_chinese_room/searle_chinese_room.html

John Searle's Chinese Room Argument

- Intelligence requires **intentional states**.
  - **intentional states:** the property of mental states being *about something* or directed at something.
- Nothing in the setup has the "right" kind of intentional states
  - Neither Searle nor the room actually understands Chinese—they're just following syntactic rules
  - Searle has intentional states, but not the right kind.
- By analogy, a computer running a program for Chinese is doing the same thing as Searle:
  - it is following formal rules to process symbol strings, but it has **no real understanding** of the language or the conversation.
  - Therefore, running an appropriate program is **not sufficient** for true understanding or a mind. Even if a computer perfectly mimics intelligent behavior, it doesn't necessarily have genuine understanding or consciousness.
- Conclusion: Strong AI is impossible

Essential argument: **Syntax** (symbol manipulation) doesn't equal **semantics** (meaning and understanding).

Philosophical objections to the CRA

1. The Systems Reply
    a. While the individual person in the room doesn't understand Chinese, the entire system (person + rulebook + room) does understand Chinese, just as individual neurons don't understand language but the brain as a whole does.
2. The Robot Reply
    a. A computer connected to sensors and actuators that interacts with the real world (embodied AI) would have genuine understanding because it could connect symbols to real-world referents, unlike the isolated room.
3. Other Minds Problem
    a. We can't truly know if anyone else understands—we only observe behavior. If we accept that other humans understand based on their behavior, why not accept that a computer passing the same behavioral tests understands?
4. Begging the Question
    a. Searle assumes from the start that following rules mechanically can't produce understanding, which is precisely what's being debated. The thought experiment may simply reflect our intuitions rather than proving anything about what understanding requires.
5. There are more…